Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Given that $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$, find $\frac{z_1}{z_2}$ in the form $a + bi$.

Multiply by conjugate: $\frac{(3+4i)(1+2i)}{(1-2i)(1+2i)} = \frac{3+6i+4i-8}{1+4} = \frac{-5+10i}{5} = -1 + 2i$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.

$\ln y = \ln x^2 + \ln 3 = \ln(3x^2) \Rightarrow y = 3x^2$. Parabola through origin, scaled by 3.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \cdots = 1 + 2x + 2x^2 + \cdots$. Valid for all $x \in \mathbb{R}$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A bag contains 6 yellow balls, 3 blue balls, and 5 red balls. A ball is drawn at random. Find $P(\text{red})$.

Final Answer: $\frac{5}{14}$. Working through this statistics step: $P(\text{red}) = \frac{5}{14} = \frac{5}{14}$.

7 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A bag contains 5 blue balls, 4 yellow balls, and 8 red balls. A ball is drawn at random. Find $P(\text{red})$.

Final Answer: $\frac{8}{17}$. Working through this statistics step: $P(\text{red}) = \frac{8}{17} = \frac{8}{17}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 8 blue balls, 6 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{red})$.

Final Answer: $\frac{6}{19}$. Working through this statistics step: $P(\text{red}) = \frac{6}{19} = \frac{6}{19}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

Final Answer: $\frac{1}{3}$. Working through this statistics step: $P(\text{blue}) = \frac{3}{9} = \frac{1}{3}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

Final Answer: $\frac{1}{4}$. Working through this statistics step: $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 3 red balls, 3 green balls, and 2 blue balls. A ball is drawn at random. Find $P(\text{blue})$.

Final Answer: $\frac{1}{4}$. Working through this statistics step: $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A bag contains 8 blue balls, 4 red balls, and 2 green balls. A ball is drawn at random. Find $P(\text{red})$.

Final Answer: $\frac{2}{7}$. Working through this statistics step: $P(\text{red}) = \frac{4}{14} = \frac{2}{7}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The data set $1,\ -1, 3, 2, 6, 10, x$ has mean 5. Find x.

Final Answer: 14. Working through this statistics step: Sum $= (5)(7) = 35 \Rightarrow x = 35 - 21 = 14$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a survey, 13 males and 6 females preferred option X, while 15 males and 14 females preferred option Y. Given that a randomly chosen person preferred option X, find the probability that the person is male.

Final Answer: $\frac{13}{19}$. $P(\text{male} \mid X) = \frac{13}{13+6} = \frac{13}{19}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Graphs & Transformations

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$. .

Final Answer: 0.65. $P(A \cup B) = 0.3 + 0.5 - 0.3 \times 0.5 = 0.65$

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Coordinate Geometry

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

Final Answer: 3360. Treat SSS as 1 item. Items: [SSS], T, A, T, I, T, I, C (8 items, 3 T's, 2 I's): 8!/(3!2!) = 3360.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Trigonometry

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

Final Answer: 10\%. By definition, $P(\text{Type I error}) = \alpha = 10\%$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Differentiation & Integration

Consider the function $f(x) = x^3 - 6x^2 + 5$. The graph of $y = f(x)$ is shown below. Determine the coordinates of the local maximum of $f(x)$.

Final Answer: (0, 5). To find the local maximum of the function $f(x) = x^3 - 6x^2 + 5$, we first need to find the critical points by taking the derivative and setting it to zero. 1. Find the first derivative: $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$ 2. Set the derivative to zero and solve for x: $3x^2 - 12x = 0$ $3x(x - 4) = 0$ This gives critical points at $x = 0$ and $x = 4$. 3. Use the second derivative test to determine if these points are local maxima or minima. Find the second derivative.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Functions: Composite & Inverse

Consider the function $f(x) = \frac{1}{x-1}$ and the function $g(x) = x^2$. Let $h(x) = f(g(x))$. The graph of $y = h(x)$ is shown below. Determine the equation of the vertical asymptote(s) of $y = h(x)$.

Final Answer: x = -1. The function $h(x)$ is a composite function, where $h(x) = f(g(x))$. We are given $f(x) = \frac{1}{x-1}$ and $g(x) = x^2$. Substituting $g(x)$ into $f(x)$, we get $h(x) = f(x^2) = \frac{1}{x^2-1}$. To find the vertical asymptotes of $y = h(x)$, we need to find the values of x for which the denominator is zero and the numerator is non-zero. The denominator $x^2 - 1 = 0$ when $x^2 = 1$, which means $x = 1$ or $x = -1$. The numerator is a constant 1, which is never zero. Therefore, the vertical.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Circular Measure

The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of a.

Final Answer: 1. Since the graph of $f(x) = ax^2 + bx + c$ passes through the given points, we can form a system of linear equations by substituting the coordinates of these points into the function's equation: 1. For $(1, -4)$: $a(1)^2 + b(1) + c = -4 \Rightarrow a + b + c = -4$ (Equation 1) 2. For $(2, -5)$: $a(2)^2 + b(2) + c = -5 \Rightarrow 4a + 2b + c = -5$ (Equation 2) 3. For $(4, -1)$: $a(4)^2 + b(4) + c = -1 \Rightarrow 16a + 4b + c = -1$ (Equation 3) Now, we solve this system of equations. Subtract Equati.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73 ✓

b. 5.21

c. 4.12

d. 3.85

Squared deviations: $(4-10)^2 = 36, (7-10)^2 = 9, (9-10)^2 = 1, (12-10)^2 = 4, (18-10)^2 = 64$. Sum $= 114$. Variance $= 114/5 = 22.8$. $\sigma = \sqrt{22.8} \approx 4.73$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. **20** ✓

b. 16

c. 24

d. 12

Mean $\bar{x} = 240/20 = 12$. Variance $= \frac{\sum x^2}{n} - \bar{x}^2 = \frac{3280}{20} - 12^2 = 164 - 144 = 20$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times \text{IQR}$ rule, what is the boundary for outliers at the upper end?

a. 69 ✓

b. 60

c. 65

d. 72

$\text{IQR} = Q_3 - Q_1 = 42 - 24 = 18$. Upper boundary $= Q_3 + 1.5 \times \text{IQR} = 42 + 1.5(18) = 42 + 27 = 69$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70 ✓

b. 0.90

c. 0.20

d. 0.65

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60 ✓

b. 0.50

c. 0.30

d. 0.75

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$ ✓

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

$P(R_1) = 4/10 = 2/5$. $P(R_2 \mid R_1) = 3/9 = 1/3$. Combined probability $= \frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$ ✓

b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

Sum of probabilities must equal 1: $k(1 + 2 + 3 + 4) = 1 \Rightarrow 10k = 1 \Rightarrow k = 1/10$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the expectation $E(X)$ for the distribution where $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$.

a. 2.1 ✓

b. 2.0

c. 2.3

d. 1.9

$E(X) = \sum xP(X = x) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For a discrete random variable X with $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

a. 9 ✓

b. 5

c. 16

d. 21

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A random variable $X \sim B(8, 0.25)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.3115 ✓

b. 0.2076

c. 0.2670

d. 0.1038

$P(X = 2) = \binom{8}{2}(0.25)^2(0.75)^6 = 28 \times 0.0625 \times 0.1779785 \approx 0.3115$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. **Mean** $= 40$**, Variance** $= 24$ ✓

b. Mean $= 40$, Variance $= 16$

c. Mean $= 50$, Variance $= 25$

d. Mean $= 40$, Variance $= 4.9$

Mean $\mu = np = 100(0.4) = 40$. Variance $\sigma^2 = np(1-p) = 100(0.4)(0.6) = 24$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$ ✓

b. 0.2500

c. 0.0879

d. 0.1563

$P(X \geq 3) = P(3) + P(4) + P(5) = \binom{5}{3}(1/4)^3(3/4)^2 + \binom{5}{4}(1/4)^4(3/4) + \binom{5}{5}(1/4)^5 = (10 \times 9 + 5 \times 3 + 1)/1024 = (90 + 15 + 1)/1024 = 106/1024 = 53/512 \approx 0.1035$ ▪

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

a. $\frac{25}{2}$

b. $\frac{240}{19}$ ✓

c. 7

d. 8

Working through this statistics step: Combined mean $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{(20)(15) + (18)(10)}{20 + 18} = \frac{240}{19}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 7 red and 8 blue balls. A ball is drawn, replaced, then a second ball is drawn. Find $P(\text{both red})$.

a. $\frac{49}{225}$ ✓

b. $\frac{112}{225}$

c. $\frac{8}{15}$

d. $\frac{7}{15}$

Working through this statistics step: $P(\text{both red}) = \left(\frac{7}{15}\right)^2 = \frac{49}{225}$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 4 blue balls, 8 yellow balls, and 3 green balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{8}{15}$ ✓

b. $\frac{1}{5}$

c. $\frac{7}{15}$

d. $\frac{4}{15}$

Working through this statistics step: $P(\text{yellow}) = \frac{8}{15} = \frac{8}{15}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 3 red balls, 5 yellow balls, and 8 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{11}{16}$

b. $\frac{1}{2}$

c. $\frac{5}{16}$ ✓

d. $\frac{3}{16}$

Working through this statistics step: $P(\text{yellow}) = \frac{5}{16} = \frac{5}{16}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 4 red balls, 3 green balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{11}$ ✓

b. $\frac{2}{11}$

c. $\frac{4}{11}$

d. $\frac{8}{11}$

Working through this statistics step: $P(\text{green}) = \frac{3}{11} = \frac{3}{11}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Differentiation & Integration

A bag contains 3 green balls, 7 red balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{2}{7}$

b. $\frac{1}{2}$

c. $\frac{11}{14}$

d. $\frac{3}{14}$ ✓

Working through this statistics step: $P(\text{green}) = \frac{3}{14} = \frac{3}{14}$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A bag contains 4 yellow balls, 6 blue balls, and 3 green balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{6}{13}$

b. $\frac{3}{13}$ ✓

c. $\frac{4}{13}$

d. $\frac{10}{13}$

Working through this statistics step: $P(\text{green}) = \frac{3}{13} = \frac{3}{13}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Circular Measure

A bag contains 8 yellow balls, 8 red balls, and 4 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{9}{20}$

b. $\frac{2}{5}$ ✓

c. $\frac{3}{5}$

d. $\frac{1}{5}$

Working through this statistics step: $P(\text{yellow}) = \frac{8}{20} = \frac{2}{5}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A bag contains 6 yellow balls, 2 green balls, and 3 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{9}{11}$

b. $\frac{6}{11}$

c. $\frac{3}{11}$

d. $\frac{2}{11}$ ✓

Working through this statistics step: $P(\text{green}) = \frac{2}{11} = \frac{2}{11}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Series & Binomial Theorem

A bag contains 3 blue balls, 7 green balls, and 2 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{1}{4}$

b. $\frac{7}{12}$ ✓

c. $\frac{1}{6}$

d. $\frac{5}{12}$

Working through this statistics step: $P(\text{green}) = \frac{7}{12} = \frac{7}{12}$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Kinematics

A bag contains 6 yellow balls, 4 green balls, and 3 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{13}$

b. $\frac{9}{13}$

c. $\frac{4}{13}$ ✓

d. $\frac{6}{13}$

Working through this statistics step: $P(\text{green}) = \frac{4}{13} = \frac{4}{13}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium

A bag contains 8 green balls, 7 yellow balls, and 6 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{13}{21}$

b. $\frac{2}{7}$

c. $\frac{1}{3}$

d. $\frac{8}{21}$ ✓

Working through this statistics step: $P(\text{green}) = \frac{8}{21} = \frac{8}{21}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 13.01 Newton's Laws

A bag contains 5 blue balls, 5 green balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{12}$

b. $\frac{1}{2}$

c. $\frac{5}{12}$ ✓

d. $\frac{1}{6}$

Working through this statistics step: $P(\text{blue}) = \frac{5}{12} = \frac{5}{12}$.

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